

JPE Replication Report

2026-05-02

1 Paper Identification

ID	20180648
Author	Tincani
Title	Peer Effects and Rank Concerns in the Classroom
Iteration	2

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
 - ☒ Details (as necessary)
- ☒ List of tables and programs

- ☐ References (Optional, but recommended)

Note: On controlled randomness. The master do-file sets set seeds for replicability.

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

Note: The data uses survey data, but it is not generated by the authors.

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

Replicator's notes (round 2):

- **Preservation duration:** Addressed. README §3.1.1 now specifies preservation for at least five years.
- **Affirmation of support for replication checks:** Addressed. README §3.1.1 now includes the explicit support commitment.
- **Codebook:** Addressed. The author added codebooks (and original questionnaires where codebooks were not available) under data/codebooks/. Mapped to datasets in README §3.2.1.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
Rendimiento por Estudiante (Ministerio de Educación; 6 CSVs, 2005–2011)	no*	yes	yes	yes	yes	yes
Matrícula por Estudiante 2009 (Ministerio de Educación)	yes	no	yes	yes	yes	yes
CPV2002 Census: PERSONA + Vivienda (INE Chile)	yes	no	yes	yes	yes	yes
SIMCE Puntajes por Estudiante (Agencia de Calidad; 4 files)	no	yes	yes	yes	yes	yes
SIMCE Cuestionarios Estudiantes (Agencia; 2 files)	no	yes	yes	yes	yes	yes
SIMCE Cuestionarios Apoderados (Agencia; 4 files)	no	yes	yes	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
SIMCE Cuestionarios Docentes (Agencia; 4 files)	no	yes	yes	yes	yes	yes
SEP School Expenditure Reports — SS_Rendidos (Ministerio)	no	yes	no	yes	yes	yes
GSHHG shoreline (NOAA; .shp + .shx)	yes	no	yes	yes	yes	yes
USGS earthquake catalog (earthquake_catalog_6.csv)	yes	no	yes	yes	yes	yes
Natural Earth 110m Admin-0 Countries	yes	no	yes	yes	yes	yes
Schools closed/receiving — EE_cerr_201011.csv (author)	yes	no	N/A	yes	yes	yes
Geocoded municipality coords — comuna_alu_geocoded.csv (author)	yes	no	N/A	yes	yes	yes
School/student-to-as-perity distances — sch_com_dist.csv, stud_com_dist.csv (author)	yes	no	N/A	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
Aggregated expenditure categories — expcat_consulting.csv, expcat_other.csv (author)	yes	no	N/A	yes	yes	yes

Note (round 2): Addressed. The author’s letter and updated README now correctly describe the Rendimiento files as restricted-access subject to Ministry approval, consistent with their placement in confidential-data-not-for-publication/raw/.

5.1.a Rendimiento por Estudiante (Ministerio de Educación, Chile)

Files: 20130425_Rendimiento_2005_20060316_PUBL.csv, 20130226_Rendimiento_2007_20080312_PUBL.csv, Rendimiento por estudiante 2008.csv, 20130227_Rendimiento_2009_20100715_PUBL.csv, Rendimiento por estudiante 2010.csv, 20130301_Rendimiento_2011_20120416_PUBL.csv. All located in confidential-data-not-for-publication/raw/.

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works [NOT click-verified]: <https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/>
- ☒ Access conditions are described:
 - Per README Note (1): dataset is publicly available on the Ministry portal, but the analysis uses specific snapshots (2005/2007/2009/2011 accessed 2013; 2010 accessed 2018; 2008 accessed 2021), and subsequent portal updates may differ.
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript (referenced as “enrollment and grade registries (Rendimiento)” in the data section)
 - ☒ in the README:

Ministerio de Educación, Chile. n.d. *Rendimiento por Estudiante, 2005–2011* [dataset]. Datos Abiertos MINEDUC. Years 2005, 2007, 2009, 2011 downloaded 2013; year 2010 downloaded 2018; year 2008 downloaded 2021. <https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/>

Note: The README describes this dataset as “publicly available” but ships it only in the confidential (privately-shared) folder rather than in data/raw/.

5.1.b SEP School Expenditure Reports (SS_Rendidos)

Files: SS_Rendidos_2008-2010_20120704_redacted.csv
and SS_Rendidos_2008-2010_20120704_redacted.dta in confidential-data-not-for-publication/raw/.

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided — no URL or DOI is given; the README only states the dataset is a restricted administrative dataset provided by Ministerio de Educación.
- ☒ Access conditions are described:
 - Restricted; access subject to Ministry approval. Per README Note (5), the version shared with the JPE Data Editor is redacted of one variable (written descriptions of single expenditure items) as per an approved exemption.
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript (school spending / SEP funds referenced in the results and mechanisms sections)
 - ☒ in the README:

Ministerio de Educación, Chile. n.d. *SEP School Expenditure Reports (SS_Rendidos_2008–2010_20120704)* [dataset]. Ministerio de Educación. Restricted administrative data; accessed 2012; access subject to Ministry approval.

5.1.c CPV2002 Census (INE Chile)

Files: CPV2002_PERSONA.csv, CPV2002_Vivienda.csv in data/raw/.

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works [NOT click-verified]: <https://www.ine.gob.cl/>
- ☒ Access conditions are described:
 - “Public microdata; official access via INE application process.” Author downloaded in 2020 and 2021.
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript (referenced as “2002 census” from the Chilean National Statistics Institute)
 - ☒ in the README:

Instituto Nacional de Estadísticas (INE), Chile. 2002. *Censo de Población y Vivienda 2002: Microdatos (Personas y Viviendas)* [dataset]. Instituto Nacional de Estadísticas. Access via official INE application process; accessed 2020 and 2021. <https://www.ine.gob.cl/>

5.1.d SIMCE datasets (Puntajes, Cuestionarios Estudiantes/Apoderados/Docentes)

All SIMCE datasets share the same access URL: <https://informacionestadistica.agenciaeducacion.cl/#/bases>

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works [NOT click-verified]: <https://informacionestadistica.agenciaeducacion.cl/#/bases>
- ☒ Access conditions are described:
 - “Restricted; access subject to Agency approval.”
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript (SIMCE test scores and questionnaires referenced throughout)
 - ☒ in the README (four separate citation entries under Section 3.3)

5.1.e USGS earthquake catalog

File: `data/raw/earthquake_catalog_6.csv`.

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works [NOT click-verified]: <https://earthquake.usgs.gov/earthquakes/search/>
- ☒ Access conditions are described:
 - Publicly available. Query: magnitude ≥ 6 , worldwide, 1900–September 2023; downloaded 9 September 2023.
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript
 - ☒ in the README:

United States Geological Survey. n.d. *USGS Earthquake Catalog (Magnitude ≥ 6 , Worldwide, 1900–September 2023)* [dataset]. USGS. Downloaded 9 September 2023. <https://earthquake.usgs.gov/earthquakes/search/>

5.1.f Author-constructed datasets

Files: `EE_cerr_201011.csv`, `comuna_alu_geocoded.csv`, `sch_com_dist.csv`, `stud_com_dist.csv`, `expcat_consulting.csv`, `expcat_other.csv` (all in `data/raw/`).

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided — N/A (included with the package)
- ☒ Access conditions are described:
 - “Included in replication package.” `EE_cerr_201011.csv` is derived from restricted Ministerio de Educación records.

- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript — round 2: all four datasets now cited. EE_cerr_201011.csv referenced in the data section; sch_com_dist.csv / stud_com_dist.csv cited via \cite{tincani_distance_measures}; comuna_alu_geocoded.csv cited in notes to Table 1; expcat_consulting.csv / expcat_other.csv cited in notes to Figure 10.
 - ☒ in the README (four author-constructed entries under Section 3.3)

5.1.g Public geographic data (GSHHG, Natural Earth)

Files: data/raw/GSHHS_f_L1.shp, data/raw/GSHHS_f_L1.shx, data/raw/ne_110m_admin_0_countries.zip.

- ☒ Dataset is provided in public deposit
- ☐ Dataset is PRIVATELY provided (not for publication)
- ☒ DOI or URL is provided, and works [NOT click-verified]:
 - GSHHG: <https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/>
 - Natural Earth: https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip
- ☒ Access conditions are described:
 - Publicly available. GSHHG downloaded 28 March 2025; Natural Earth downloaded 24 February 2026.
- ☐ Data are not cited, but a working paper, article, or other document is cited
- ☒ Data are cited
 - ☒ in the manuscript — round 2: GSHHG cited in notes to Table 1, Natural Earth cited in notes to Figure 7.
 - ☒ in the README:

National Centers for Environmental Information (NOAA). n.d. *Global Self-consistent Hierarchical High-resolution Geography (GSHHG)*, latest version [dataset]. NOAA. Downloaded 28 March 2025. <https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/>

Natural Earth. n.d. *Natural Earth 1:110m Cultural Vectors – Admin 0 Countries* [dataset]. Natural Earth. Downloaded 24 February 2026. https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

Note (round 2): Both addressed. The README title was updated to conform to JPE guidance and the author’s affiliations were added.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - **Operating System:** README Section 4.1 states the package was tested on **Windows 10 only**. No testing on macOS or Linux is reported.
 - **Python 3.12** (tested 3.12.10). Required packages with versions:
 - pandas==2.3.3, numpy==2.3.4, geopandas==1.1.1, pyogrio==0.11.1, pyproj==3.7.2, shapely==2.1.2, geopy==2.4.1, matplotlib==3.10.7, geodatasets==2024.8.0
 - Plus transitive dependencies listed explicitly in the README.
 - **Stata:**
 - **StataNow 19.5** for all do-files except one.
 - **Stata 15** for 21_conley.do only.
 - All do-files specify version 18, except 21_conley.do (version 15) and 9_lca_all.do (version 15 for gsem).
 - **R 4.5.2** (2025-10-31 ucrt). Packages: ggplot2 4.0.2, dplyr 1.2.0.
- ☒ Computational Requirements specified as follows:
 - **Local (main pipeline):** Intel Core Ultra 7 268V 2.20 GHz, 32 GB RAM (31.6 GB usable), 64-bit x64-based processor.
 - **Cluster (Conley only):** UCL Myriad HPC. Modules: beta-modules, gcc-libs/9.2.0, libpng/1.6.37/gnu-9.2.0, stata/15, plus gerun, emacs/28.1, userscripts/1.5.0, rcps-core/1.0.0, mpi/intel/2018/update3/intel. Sun Grid Engine resource requests: 188 GB memory, 300 GB scratch, 1 OpenMP thread.
 - **Memory:** Main pipeline runs with 32 GB RAM locally. Conley estimation requires ~188 GB (cluster-scale).
- ☒ Time Requirements specified as follows:
 - Full Stata pipeline excluding Conley: **~7 hours** locally.
 - 21_conley.do on UCL Myriad: **~61 hours**.
 - calculate_coastal_proximity.py: < 10 minutes.
 - create_map.py: < 1 minute.
 - plot_histogram_earthquakes.R: < 1 minute.
- ☒ Requirements are complete.

Note (round 2): The author also added codebooks (and original questionnaires where codebooks were not available) under data/codebooks/, mapped to datasets in README §3.2.1.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 12 - Variables flagged in data: 55 - Code files with PII references: 42 - PII references in code: 4586

7.1.a Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	census_parents_clean_2023.dta	5	father, mother, degree, child, school, house
Data	census_parents_clean_wclass_all.dta	5	father, mother, degree, child, school, house
Data	census_with_class_type_true_predicted_all.dta	5	father, mother, degree, child, school, house
Data	comuna_all_geocoded.dta	0	lat, lon
Data	data_final2for_regs_all.dta	12	school, father, mother, house, loc
Data	data_final2for_regs_teach_all.dta	12	school, father, mother, house, loc
Data	data_for_map_2025_all.csv	0	lat, lon
Data	donut_graph_construction_types.xlsx	0	census, name
Data	eb_map_iv1f.txt	1	name
Data	post_for_predict_2025.dta	4	gender, father, mother, house
Data	pre_for_predict_2025.dta	4	gender, father, mother, house
Data	towns_with_distance_to_coast.csv	0	lat, lon
Code	0_Peer_Effects_and_Rank_Concerns.do	8	name, loc, school, census, minute, lat
Code	10_predict_vulnerability.do	21	census, son, dob, second, name
Code	11_geog_coastal_proximity.do	0	school, house, name, loc
Code	12_gen_variables.do	9	lat, dob, son, name, school, loc, lon
Code	13_gen_data_for_map.do	7	name, lat, lon, coord
Code	14_analyze_census.do	38	house, lat, child, school, census, name, loc, dob, father, lon, mother

File Type	File	Variables/References	PII Categories
Code	15_descriptive_tables_figures.do	108	school, name, house, lon, loc, lname, lat, father, mother
Code	16_main_effects_and_rob.do	116	school, name, loc, lat, municipality
Code	17_het_effects.do	118	school, name, loc, lon
Code	18_med_relationships.do	130	school, name, loc, lat
Code	19_school_resources.do	156	school, sex, son, lat, loc, name, lname
Code	1_geog_codes_rendimiento.do	162	name
Code	20_identifying.do	164	school, name, loc, lname
Code	21_conley.do	180	lat, minute, son, name, lon
Code	2_clean_census.do	188	house, son, census, child, school, father, mother, wife, husband, degree, name
Code	3_clean_source_4alu_8alu.do	191	school, birth, name, second
Code	5_clean_4spad_8cpad.do	238	father, mother, house, name, second, son
Code	6_clean_source_8prof.do	212	name, school
Code	7_clean_rendimiento.do	260	school, name, country
Code	8_merge.do	25	lon, school
Code	9_lca_all.do	17	name, census, loc
Code	_gcorr.do	106	lat, loc
Code	_grmiss2.do	27	loc
Code	analysis_earthquake_effects.tex	140	lon, lat, loc, location, school, second, country
Code	appendix.tex	116	house, lat, sex, school, loc, second, lon
Code	appendix_reconstruction_policy.tex	7	school, lat, loc, location, second
Code	calculate_coastal_proximity.py	10	coord, lat, lon, zip

File Type	File	Variables/References	PII Categories
Code	conclusion.tex	25	lon, school
Code	create_map.py	9	coord, lat, lon, zip, name
Code	data.tex	8	gender, house, school, lat, country, loc, lon, second
Code	estadd.ad	736	loc, lon, name, lname, lat
Code	estout.ad	657	loc, name, son, lname, lat, block, lon
Code	estpost.ad	596	lat, loc, name, lon, lname, son
Code	eststo.ad	000	loc, lon, name
Code	esttab.ad	524	loc, lon, name, second
Code	introduction.tex	6	child, school, house, loc, lat, location, son, gender, lon
Code	manuscript.tex	1	lon, url
Code	mechanism.tex	13	school, lat, loc, location, son, second, lon, gender, social
Code	model_extension.tex	42	lon, lat
Code	ols_spatial_HAC.ado	81	lat, lon, city, degree, loc, location, coord, name, second, son, lname
Code	plot_histogram_earthquakes.R	20	name
Code	theoretical_model_rank_concerns.tex	81	lat, school, lon, son, city

See Appendix for detailed listing of all flagged instances.

7.1.b File Paths Report

Generated on 2026-04-23T16:06:01.744

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for *all* STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
44	4	16	2	0

7.1.c Duplicate Files Report

We did not find any duplicate files.

7.1.d Zero Size Files Report

We did not find any zero sized files.

7.1.e Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/CPA8VO.xlsx	138.66
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_parents_clean_2023.dta	182.68
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_parents_clean_wclass_all.dta	203.96
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/2009 8b bbdd cpadres.csv	251.71
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/CEST8VO.csv	263.16
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_with_class_type_true_predicted_all.dta	267.8
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/Rendimiento por estudiante 2010.csv	397.4

Filename	Size (MB)
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/raw/20130227_Rendimiento_2009_20100715_PUBL.csv	400.06
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/raw/ Rendimiento por estudiante 2008.csv	412.74
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/raw/20130226_Rendimiento_2007_20080312_PUBL.csv	414.26
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/raw/20130425_Rendimiento_2005_20060316_PUBL.csv	423.26
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/raw/20130301_Rendimiento_2011_20120416_PUBL.csv	423.31
/Replication_Package_Tincani-20180648/ confidential-data-not-for-publication/ processed/data_final_for_regs_all.dta	480.37
/Replication_Package_Tincani-20180648/ confiden- tial-data-not-for-publication/processed/ data_final_for_regs_teach_all.dta	485.77

7.2 Code description

cloc	github.com/AIDanial/cloc v 2.06 T=2.24 s (27.7 files/s, 947865.2 lines/s)
------	---

Language	files	blank	comment	code
CSV	15	0	0	2099429
Stata	31	3096	759	16544
TeX	12	976	26	2998
Python	2	58	34	114
Text	1	0	0	76
R	1	12	11	34
---	---	---	---	---
SUM:	62	4142	830	2119195

- **31 Stata do-files** (~16,500 lines of Stata code). Numbered 0_* through 21_*, plus 8 bundled frozen user-written ados in code/stata/ado_frozen/ (estout suite, egenmore, rmiss2, ols_spatial_HAC).
- **2 Python scripts:** calculate_coastal_proximity.py (preprocessing, Step 1) and create_map.py (post-Stata, Step 3; produces Figure 7).
- **1 R script:** plot_histogram_earthquakes.R (Step 4; produces Figure 2; self-contained and independent of the Stata pipeline).
- **1 Excel workbook** for manual verification of Figure 3 (donut graph): code/excel/donut_graph_construction_types.xlsx.
- **12 TeX files** (paper + appendix source) — not part of the reproducibility pipeline per se, but determine which outputs feed into the final compiled paper.
- ☒ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

Note: The authors provide a program-level mapping in README §11 (Figures, Tables, In-text numbers).

7.3 Computing Environment of the Replicator

The main replication was performed on a Nuvolos cloud instance provisioned by the Data Editor. Initial package setup and report-writing were performed locally on an Apple MacBook Air (M1, 8 GB RAM, macOS Sequoia 15.5), but the full computational pipeline ran on Nuvolos because of memory-concerns.

Nuvolos instance specifications:

- Linux x86_64 (conda (base) environment visible from terminal)
- 64 GB RAM
- Separate containers for the Stata, R, and Python applications.

Software used:

- **StataNow 19.5 SE** (Stata/SE, not /MP — some do-files emit set processors allowed only with Stata/MP; execution continues via capture noisily)
- **Python 3.12** (conda (base) env) with pinned packages per README §4.2: pandas 2.3.3, numpy 2.3.4, geopandas 1.1.1, geopy 2.4.1, matplotlib 3.10.7, geodatasets 2024.8.0, plus transitive dependencies
- **R 4.x** (with ggplot2 and dplyr auto-installed by the script)

Round 2: The Conley GPA + 350 km re-execution was run on a separate Nuvolos instance scaled to **128 GB RAM / 16 cores** via the paid credit tier.

8 Replication

1. Conley estimation (code/stata/21_conley.do) skipped. Per README §8.2, this do-file requires 188 GB RAM and ~61 hours of runtime on a Stata 15 HPC cluster (UCL Myriad). This exceeds the memory of both my local machine and the default Nuvolos instance. Table A14 was verified against the author's log file (see README §6.2) rather than re-executed. To skip this step, I commented out line 218 of code/stata/0_Peer_Effects_and_Rank_Concerns.do.
2. Python Step 1 (code/python/calculate_coastal_proximity.py). Installed pinned packages on the Nuvolos (base) conda environment. Script ran to completion.
3. Stata master do-file: first run blocked by path-detection bug. After cd'ing into the package root and invoking `do code/stata/0_Peer_Effects_and_Rank_Concerns.do`, the do-file's own root-detection logic immediately redirected the working directory away from the package root (to /home/datahub), causing the script to exit with `r(198)`:

```
. capture noisily cd "`=subinstr("`c(filename)'" ,  
"/0_Peer_Effects_and_Rank_Concerns.do", "", .)'"  
/home/datahub  
...  
ERROR: Stata must be started in the replication package root directory.  
r(198);
```

Commented out line 36 of code/stata/0_Peer_Effects_and_Rank_Concerns.do, then the code ran to completion.

4. Stata master do-file: log bug. Second run from the Stata GUI with an outer `log using "nuvolos_run.log"`; data-prep do-files 1–8 ran successfully, then 9_lca_all.do aborted with `log file already open, r(604)` because eight subordinate do-files open their own default log via `log using ...`, replace without a preceding `capture log close`, which precludes any replicator-side outer log; closed the outer log and re-launched without one.
5. Stata master do-file: Four outputs (Figure 6 and Tables 4, A6, A12) and the second panel of Table 3 were initially missing because of 5 Windows-style separators in 15_descriptive_tables_figures.do (L582), 16_main_effects_and_rob.do (L197, L271), and 20_identifying.do (L266, L303); the replicator patched them to /, re-ran the 3 affected do-files, and confirmed all outputs were then produced.
6. Python Step 3 (code/python/create_map.py). Ran to completion
7. R Step 4 (code/R/plot_histogram_earthquakes.R). The script's own root-directory auto-detection is broken for any invocation mode other than `Rscript ...` from the command line. After working around this by hard-coding the working directory, the script ran to completion.
8. Round 2 verification. Author's reply letter (23 April 2026) addressed all round 1 flagged issues; spot-checked: master do-file line 36 commented out, `capture log close` precedes every `log using` in the eight subordinate do-files, no Windows-style \ separators remain in output paths, README updated with title + affiliations + preservation/support statement + codebooks.

Conley GPA + 350 km independently re-executed on Nuvolos (16 cores / 128 GB RAM, ~5h 14min wall clock); the three lead Conley standard errors match Table A14, Panel B, Column (5) of the bundled manuscript PDF exactly: *post_st_exp* SE 0.0056 (), *post_loo_mean_damage* SE 0.0134 (), ***post_loo_sd_damage* SE 0.0112 (*)**. Other Conley distance bandwidths (10 / 25 / 50 km) and Column (5) of Panel A (test scores at 250 km) were not independently re-run, per the Data Editor’s instruction.

9 Findings

The replication package **fully reproduces** the author’s bundled manuscript PDF (replication-package/Replication_Package_Tincani-20180648/Manuscript_Replication_Package_Tincani-20180648.pdf), dated March 2026 and shipped with the replication deposit. Every figure, table, and in-text number produced by the replicator’s run matches the corresponding exhibit in that PDF. See package-output-map.xlsx for a full list of reproduced outputs.

Assumption about the authoritative PDF. Per the replicator’s correspondence with the Data Editor, the March 2026 Manuscript_Replication_Package_Tincani-20180648.pdf is treated as the authoritative version for this replication. The replicator also compared a selection of outputs against the earlier paper-appendices/20251210_Peer_Effects_Tincani.pdf (dated 10 December 2025) that was delivered with the assignment email; this older snapshot differs from the bundled PDF by a handful of observation counts. This suggests that the author re-ran the pipeline between the December 2025 submission and the March 2026 replication package.

Summary of code issues that required replicator intervention in round 1 (all addressed by the author in round 2 — verified):

1. The master do-file’s `c(filename)`-based `cd` auto-detect fails on Linux/Nuvolos (redirects to /home/datahub, outside the package root). Round 2: author commented out line 36 of `0_Peer_Effects_and_Rank_Concerns.do`.
2. The R script’s `commandArgs()`-based root-directory auto-detect fails when the script is `source()`d from an R console rather than invoked via `Rscript` from the shell. Round 2: author added a note in the README documenting the required invocation mode.
3. Five Windows-style `\` path separators in subordinate Stata do-files caused four output files to be missing and one appended table to be incomplete on Linux. Round 2: author replaced all `\` with `/` in output paths.
4. All subordinate Stata do-files that use `log using` did so without a preceding `capture log close`, precluding any replicator-side outer `log`. Round 2: author added `capture log close` before every `log using` in the eight affected do-files.

9.1 Data Preparation Code

All 13 Stata data-preparation do-files (1_... through 13_...) and the preparatory Python script (`calculate_coastal_proximity.py`) ran to completion and produced the expected datasets in `confidential-data-not-for-publication/processed/`.

9.2 Tables and Figures

Data Prepara- tion Pro- gram	Dataset created	Reproduced?
	code/stata/1 geog_regides_rendimiento.do geog_regides_rendimiento.do	Yes
	code/stata/2 casus_parents_clean_2023.dta casus_parents_clean_2023.dta (+ CPV2002_PERSONA/ Vivienda.dta)	Yes
	code/stata/3 clean_4in2005/2007/2009/2011.dta clean_4in2005/2007/2009/2011.dta baseline_simce_2005	Yes
	code/stata/4 clean_simc8o82009/2011_cleaned.dta clean_simc8o82009/2011_cleaned.dta	Yes
	code/stata/5 baseline_cpall8cpindo_cpap_2005/2007/2009/2011.dta baseline_cpall8cpindo_cpap_2005/2007/2009/2011.dta	Yes
	code/stata/6 oldlinesimce8imgf.do oldlinesimce8imgf.do mate_2009/2011.dta	Yes
	code/stata/7 clean_rendim_2008/2009/2010/2011_cleaned.dta clean_rendim_2008/2009/2010/2011_cleaned.dta	Yes
	code/stata/8 imogcpad*_alucpad*_cleaned_2023.dta; imogcpad*_alucpad*_cleaned_2023.dta; pre/ post_for_predict_2025.dta	Yes
	code/stata/9 casus_parents_clean_wclass_all.dta casus_parents_clean_wclass_all.dta (LCA); output/figures/ lca_all.log (Figure 3)	Yes
	code/stata/10 predict_with_class_bility_predoe_predicted_all.dta; predict_with_class_bility_predoe_predicted_all.dta; pre/ post_for_predict_class_*.dta	Yes
	code/ towns_with_distance_to_coast.Xls python/calculate_coastal_proximity.py	Yes
	code/stata/11 geog_coastal_proximity.dta; geog_coastal_proximity.dta; towns_with_distance_to_coast.dta	Yes
	code/stata/12 lagfinalvariables_all.dta; datafinal_for_regs_teach_all.dta; lagfinalvariables_all.dta; datafinal_for_regs_teach_all.dta; prepost_cpad_alu_all.dta (main analysis dataset)	Yes
	code/stata/13 datafor2025pal.csv; datafor2025pal.csv; comuna_alu_geocoded.dta	Yes
Figure/ Table #	Program	Line Number Reproduced?

Data Prepara- tion Pro- gram	Dataset created	Reproduced?	
Figure 1	code/tex/Paper/data.tex	N/A (tikz; not gener- ated from data)	N/A (conceptual tikz)
Figure 2	code/R/plot_histogram_earthquake.R	(full script)	Yes
Figure 3	code/stata/9_lca_all.do + 9_lca_all.do line 7 code/excel/donut_graph_construction Excel.assem- bled manually		Yes (manual Excel verifica- tion per §6.1)
Figure 4	code/ stata/14_analyze_census.do	201	Yes
Figure 5	code/ stata/14_analyze_census.do	219	Yes
Figure 6	code/stata/15_descriptive_tables582figures.do		Yes (after backslash-path patch)
Figure 7	code/python/create_map.py	(full script)	Yes
Figure 8	code/stata/17_het_effects.do	628	Yes
Figure 9	code/stata/17_het_effects.do	875	Yes
Figure 10	code/stata/19_school_resources479		Yes
Figure 11	code/stata/18_med_relationship341b		Yes
Figure 12	code/tex/Paper/	N/A (conceptual drawing)	N/A (conceptual drawing)
Figure A1	code/tex/Paper/ reconstruccion_dios.png	N/A (field photo- graph)	N/A (field photograph)
Figure A2	code/stata/17_het_effects.do	1,102	Yes
Figure A3	code/stata/18_med_relationship395b		Yes
Figure A4	code/stata/15_descriptive_tables598figures.do		Yes
Table 1	code/stata/15_descriptive_tables261figures.do	293, 332 (three esttab calls append- ing)	Yes
Table 2	code/stata/16_main_effects_and540b.do		Yes

Data Prepara- tion Pro- gram	Dataset created	Reproduced?		
Table 3	code/stata/20_identifying.do	213		Yes (after backslash-path patch to append)
Table 4	code/stata/20_identifying.do	303		Yes (after backslash-path patch)
Table 5	code/stata/20_identifying.do	366		Yes
Table 6	code/stata/17_het_effects.do	166		Yes
Table 7	code/stata/18_med_relationships.do	115		Yes
Table 8	code/stata/19_school_resources.do	366		Yes
Table 9	code/stata/18_med_relationships.do	115		Yes
Table A1	code/tex/Paper/appendix.tex	N/A	(conceptual table)	N/A (conceptual table)
Table A2	code/ stata/14_analyze_census.do	54	(outfile); (write)	178 Yes
Table A3	code/stata/15_descriptive_tables.do	482	(outfile, con- structed via write) file	Yes
Table A4	code/stata/15_descriptive_tables.do	549	figures	Yes
Table A5	code/stata/16_main_effects_and_rob.do	97		Yes
Table A6	code/stata/16_main_effects_and_rob.do	97		Yes (after backslash-path patch)
Table A7	code/stata/20_identifying.do	166		Yes
Table A8	code/stata/16_main_effects_and_rob.do	150		Yes
Table A9	code/stata/17_het_effects.do	387		Yes
Table A10	code/stata/17_het_effects.do	236		Yes
Table A11	code/stata/17_het_effects.do	282		Yes
Table A12	code/stata/16_main_effects_and_rob.do	271		Yes (after backslash-path patch)
Table A13	code/stata/16_main_effects_and_rob.do	312		Yes

Data Prepara- tion Pro- gram	Dataset created	Reproduced?	
Table A14	code/stata/21_conley.do	76 (ts); 99 (GPA); 134 (log)	Yes (round 2: Panel B Column (5) – GPA + 350 km – independently re-executed on Nuvolos and matches the manuscript exactly; other columns/panel verified against the author-provided log)
Table A15	code/stata/16_main_effects_and_35b.do	35b	Yes
Table A16	code/stata/19_school_resources.do	320	Yes
Table A17	code/stata/19_school_resources.do	259	Yes
Table A18	code/stata/19_school_resources.do	156	Yes
Table A19	code/tex/Paper/appendix_recons/N/A (external table, translated)	N/A (external table, translated)	N/A (external table, translated)
Table A20	code/tex/Paper/appendix_recons/N/A (conceptual table)	N/A (conceptual table)	N/A (conceptual table)
In-text numbers	Program	Line Number	Reproduced?
Section 2.1 (in-text)	code/ stata/12_gen_variables.do	Paper §2.1 code: 12_gen_variables.do:129 (log using)	Yes
Section 2.2 (in-text)	code/ stata/14_analyze_census.do	Paper §2.2 (intro) code: 14_analyze_census.do:21 (log using)	Yes
Sections 2.3 & 3 (in-text)	code/stata/15_descriptive_tables_figures.do	Paper §2.4 + §3 (intros) code: 15_descriptive_tables_figures.do:49 (log using)	Yes

Data Prepara- tion Pro- gram	Dataset created	Reproduced?
Section 4.1.1 (in-text)	code/stata/16_main_effects_and_rob.do:413 (log using)	Paper §4.1.1 code: 16_main_effects_and_rob.do:413 (log using)
Section 4.1.2 (in-text)	code/stata/18_med_relationships.do:136 (log using)	Paper §4.1.2 code: 18_med_relationships.do:136 (log using)
Section 4.1.2 (in-text CIs)	code/stata/18_med_relationships.do:92 (file open)	Paper §4.1.2 (95% CIs) code: 18_med_relationships.do:92 (file open)
Section 4.1.3 (in-text)	code/stata/19_school_resources.do:228 (log using)	Paper §4.1.3 (intro) code: 19_school_resources.do:228 (log using)

All 43 exhibits that are generated from data (i.e. excluding the six conceptual / external items: Figures 1, 12, A1 and Tables A1, A19, A20) were produced successfully and match the manuscript PDF. The filled package-output-map.xlsx reports Yes for every one of these, with inline notes where special treatment was required:

- **Figure 3** (donut graph): Yes (manual Excel verification per README §6.1).
- **Table A14** (Conley): Yes. Round 2: at the Data Editor’s request, the GPA + 350 km specification (Panel B, Column (5) — the most computationally intensive single cell, ~8 hours on the author’s HPC) was independently re-executed by the replicator on a Nuvolos 16-core / 128 GB instance via a wrapper extracting only that block of 21_conley.do. Wall clock: ~5h 14min. The three lead Conley standard errors and significance stars match the manuscript exactly. Other distance bandwidths (10/25/50 km) and Panel A Column (5) (test scores at 250 km) were not independently re-run, per the Data Editor’s instruction; verified against the author-provided Table_A14_conley.txt log per README §6.2.

9.3 In-Text Numbers

- ☒ All in-text numbers verified.

The six in-text-numbers log files (Section_2_1.txt, Section_2_2.txt, Sections_2_3.txt, Section_4_1_1.txt, Section_4_1_2.txt, Section_4_1_3.txt) in output/in_text_numbers/ were regenerated by the replicator’s run. Every in-text claim in the corresponding paper section was located in its log file and found to match the printed value in the bundled manuscript PDF.

The med_effects_curriculum_CIs_2025.tex file under output/tables/ (not a table in the paper but the source of the Section 4.1.2 in-text 95% confidence intervals) was also verified against the paper text.

10 Classification

- ☒ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☒ None.
- ☐ Discrepancy in output (either figures or numbers in tables or text differ)
- ☐ Bugs in code that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ Code missing, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ Data preparation code missing should be checked if the code missing seems to be data preparation code
- ☐ Code not functional is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ Software not available to replicator may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ Insufficient time available to replicator is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ Data missing is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ Data not available is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ Missing README is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

Note (round 2): The four code-level issues that required replicator intervention in round 1 (master do-file cd auto-detect, R script commandArgs() auto-detect, five Windows-style \ path separators, missing capture log close before subordinate log using calls) have all been addressed by the author in the round 2 deposit. Reason updated from Bugs in code to None.